



Module 1: Introduction to Human-AI Collaboration

Course: Human-AI Collaboration
Instructor: Dr. Christopher Flathmann



COURSE OVERVIEW

About Your Instructor – Christopher Flathmann

Professor in Human-Centered Computing

Co-Director of the Clemson University Center for Human-AI Interaction, Collaboration, and Teaming

Expert in Human-AI Teamwork

B.S. in CS from Clemson, Ph.D. in HCC from Clemson

Domain-Agnostic, I believe AI can Help Anyone if Properly Designed and Integrated





Learning Objectives

- Define human-AI collaboration vs. automation
- Trace historical milestones relevant to collaboration
- Summarize empirical findings on human+AI synergy & augmentation
- Identify major design challenges in human-AI teaming
- Understand how weekly journals and projects map to lecture material

Course & Format Overview

- Fully asynchronous: lectures, readings, journals, projects
- Focus: designing healthy human-AI systems
- Components: Research Journal, AI in the News Journal, Project 1, Project 2, midterm, final
- Learners advance independently; reflections and quizzes replace live discussion

Weekly Assignments (Purpose & Details)

- Research Journal (2 pages/week): reflect on a research article (core ideas, critique, implications)
- AI in the News Journal (1 page/week): link real AI developments to course themes (successes, failures, ethics)
- Purpose: anchor lecture content in scholarship and real-world developments

Major Projects

Project 1: Technology
Assessment —
evaluate existing AI
system

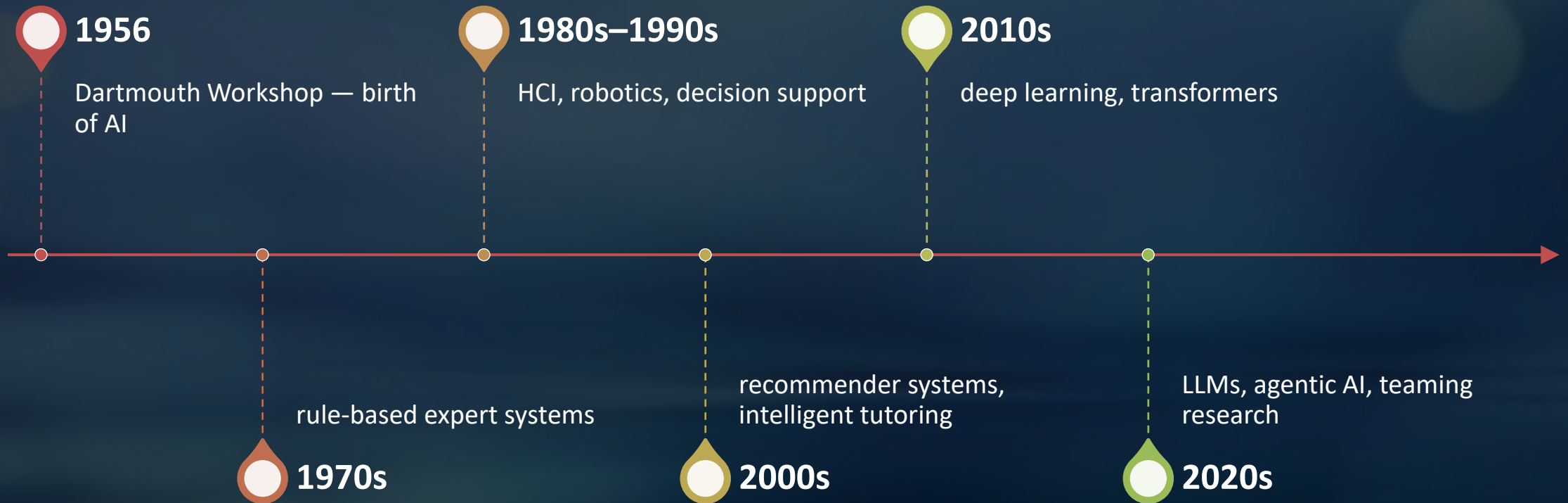
Project 2: Future AI
Vision — speculative
design + critical
analysis



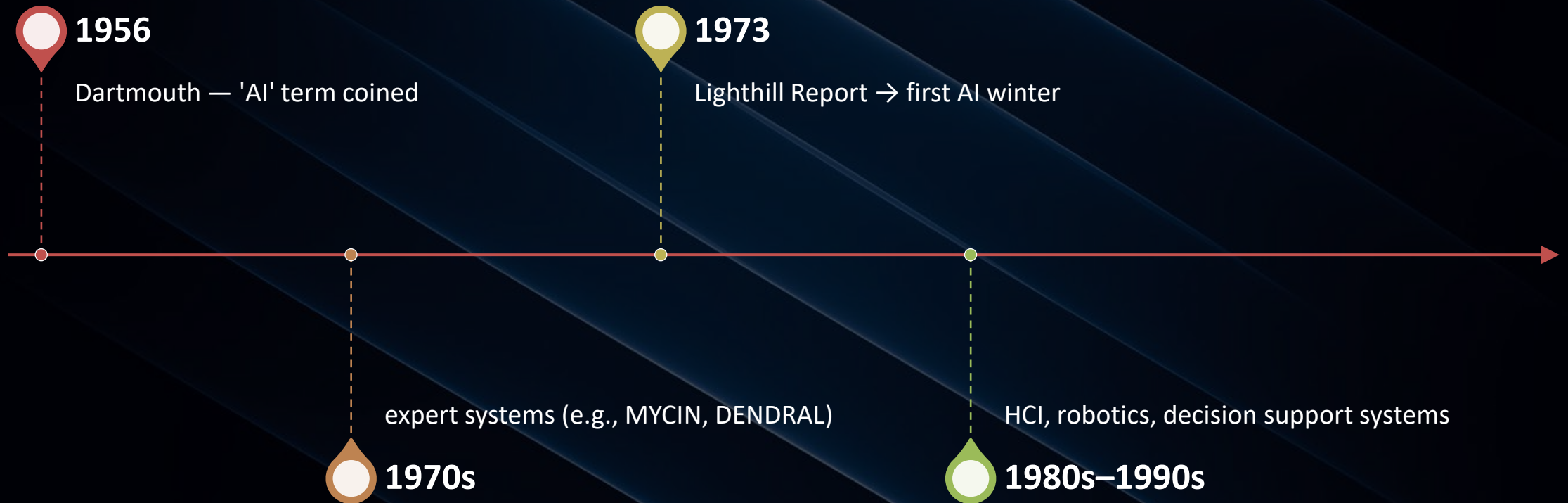
HOW DID WE GET HERE?



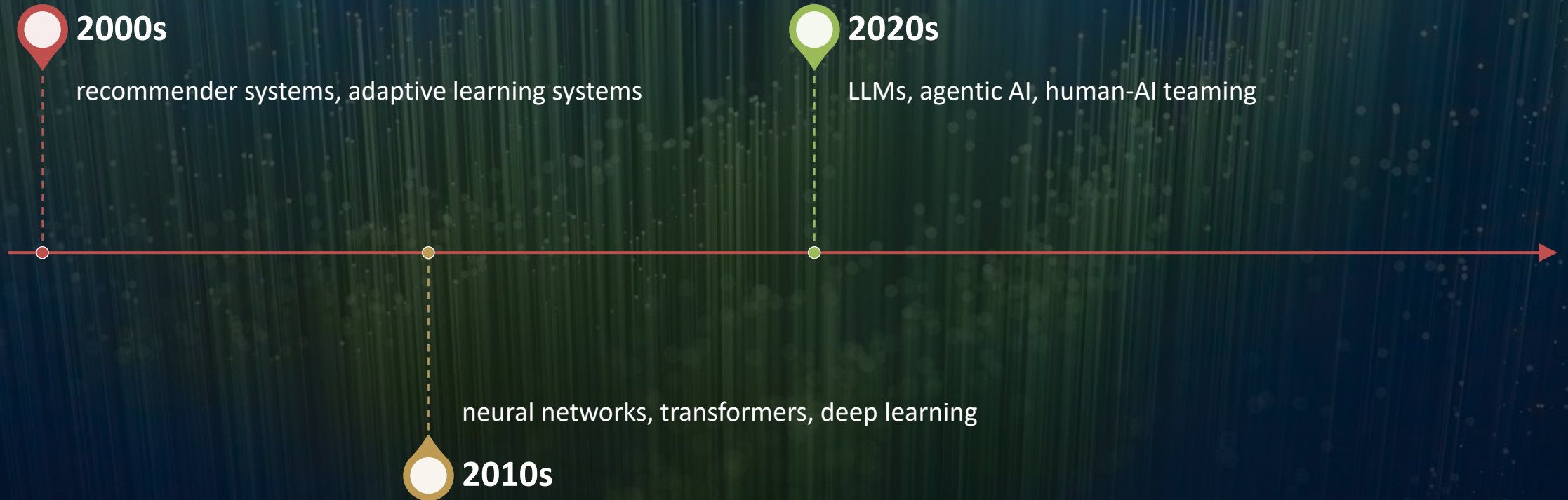
Historical Timeline: Key Milestones (Philosophy Started Way Before This)



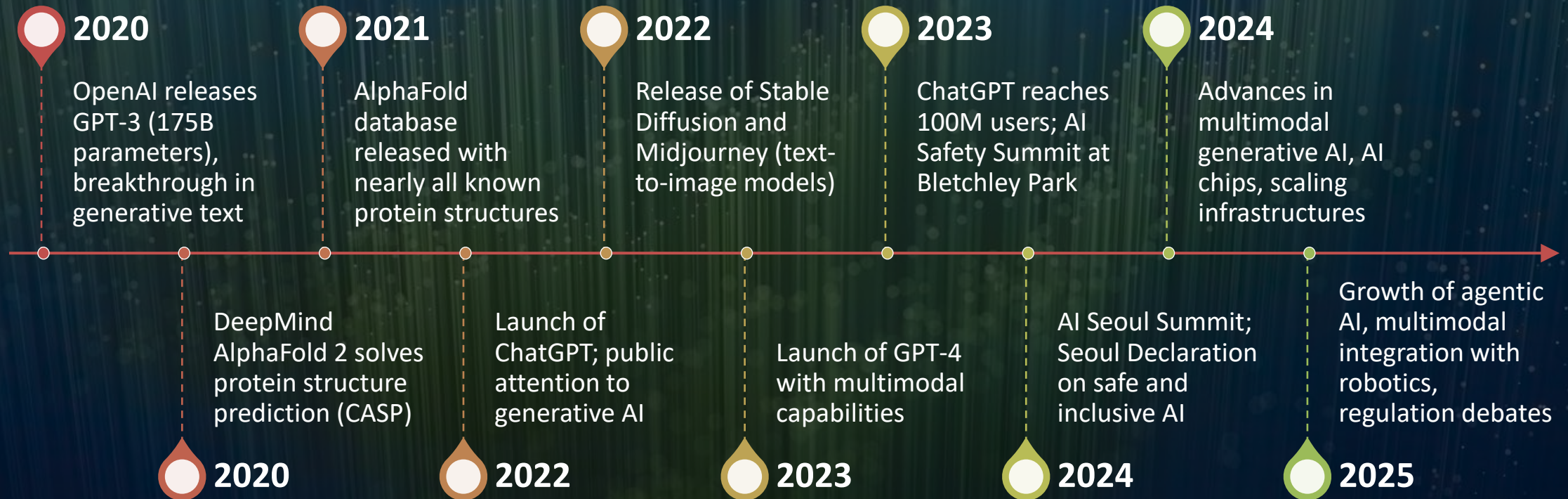
Timeline: Early to Middle Era



Timeline: Recent Era to 2020



Key AI Milestones: 2020 → Today



Timeline Projection: Future



Increasing autonomy of
AI agents



Widespread
deployment of AI
teammates



Focus: teaming
protocols, trust,
alignment



Research emphasis:
symbiosis, not control

Why Human-AI Collaboration Matters

Humans: ethics, social awareness, context reasoning



AI: speed, scale, pattern recognition



Collaboration: combining strengths, mitigating weaknesses



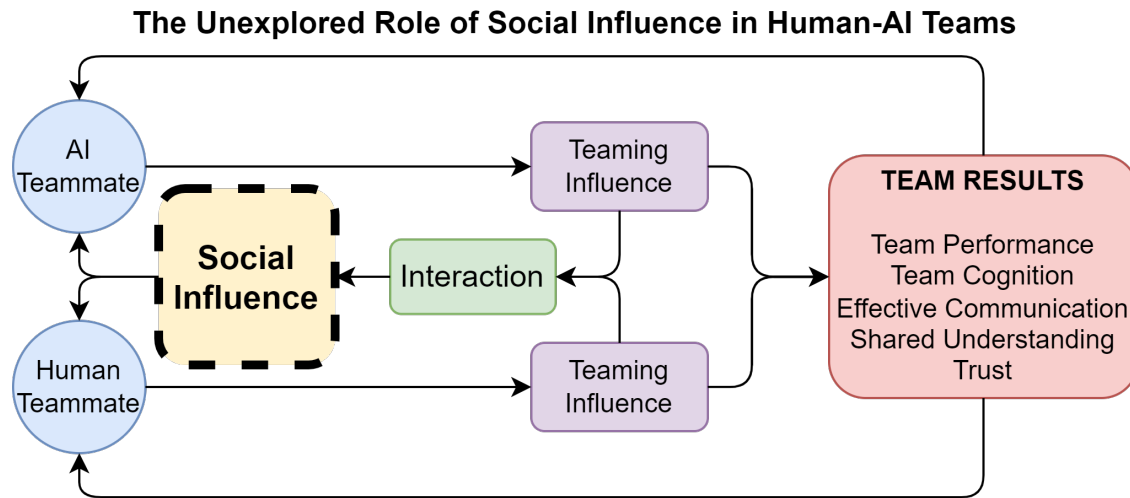
UNDERSTANDING INTERACTION, COLLABORATION, AND TEAMING



Food for Thought

- When you hear 'AI collaboration,' what systems or tools come to mind?
- Explore how your answer to this question changes throughout this course

Automation vs. Collaboration

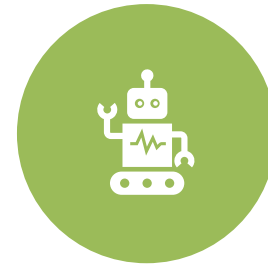


- Automation: AI acts independently, human oversight is minimal
- Collaboration / Teaming: AI and human share control, influence one another
- Goal shift: from AI as passive tool → AI as active partner
- Is Delegation Collaboration?

Spectrum of Roles: Tool → Assistant → Teammate



Tool: human leads, AI executes



Assistant: AI suggests, human decides



Teammate: bidirectional influence, role flexibility, shared goals



Most real systems blend these roles

Examples: Tool / Assistant

Grammarly: suggests edits, human finalizes

Recommender systems: propose items, human selects

Decision-support dashboards in medicine: show risk scores, human clinicians decide

Low autonomy, narrow domain

Example: Teammate Role

AI co-pilot adjusting
route/behavior

Conversational agents negotiating
tasks

Agents in strategy games
coordinating with human players

Key: AI may initiate, ask questions,
adapt

Use Case Decides What Role is Appropriate

				
Customer definition	Eyes-on Hands-on	Eyes-on Hands-off	Eyes-off Hands-off	No driver (Robotaxi)
	Driver Present			Tele-Operations
Safety MRM	Based on driver	Based on driver Driver Monitoring System needed	Full MRM capability is mandatory - Stopping safely on the shoulder of the road - Stop-in-lane is not safe enough	
Non-safety MRM (deadlocks)	Based on driver			Tele-Operations
* MRM- Minimum Risk Maneuver				



Human-Autonomy Teaming (HAT) Fundamentals

Humans & agents work
interdependently
toward shared goals

O'Neill et al. (2022):
agent traits, task design,
team dynamics,
communication
modalities

Autonomous agents
expected to coordinate,
adapt, take initiative



HAT vs Traditional Automation

- Traditional: human monitors, intervenes
- HAT: each actor contributes actions, coordinates
- Concepts: interdependence, role flexibility, shared mental models
- Many systems still think of autonomy as 'advanced automation'

Empirical Findings: Meta-Analysis (Vaccaro et al.)

Reviewed 106 experiments, 370 effect sizes

Defined human-AI synergy (combined > both solo) and human augmentation (combined > human alone)

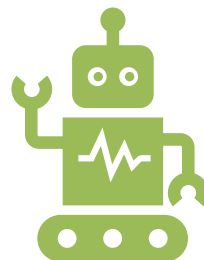
Currently: human + AI < best of human or AI alone

But often human+AI > human alone

Moderators of Collaboration Success



Task type: decision tasks often suffer; creative tasks show gains



Relative competence: if human > AI, combining more likely useful; if AI > human, could drag



Other factors (e.g. explanation, confidence scoring) show mixed results

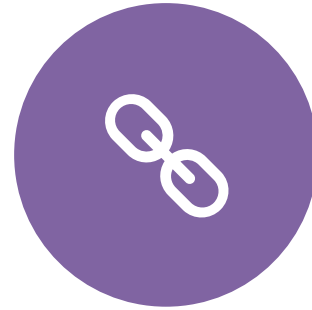
Implications from the Evidence



COLLABORATION IS NOT
AUTOMATICALLY BETTER



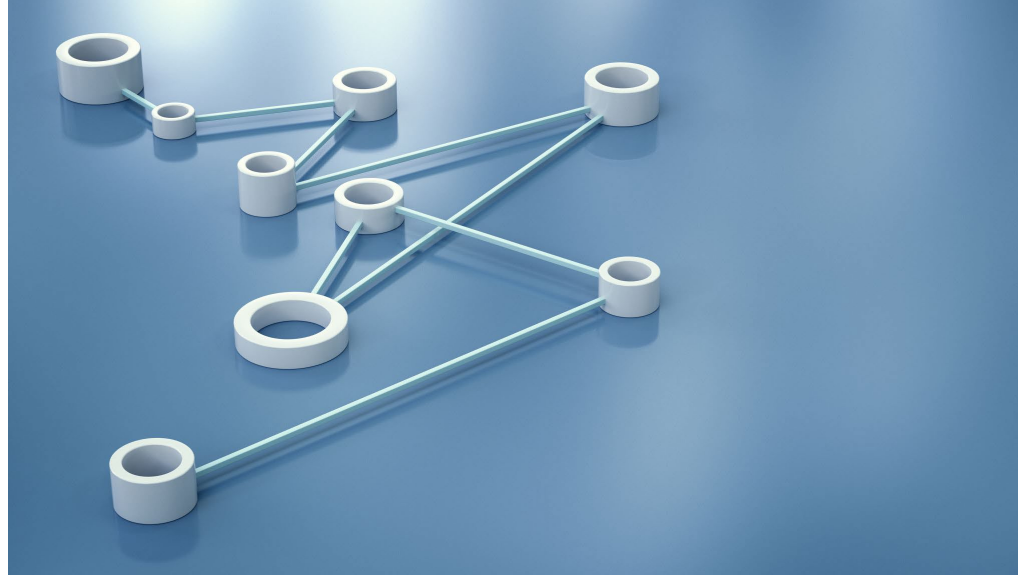
DESIGN MUST ACCOUNT
FOR TASK, ROLES,
INTERFACE, TRUST



CONTEXT-SPECIFIC: AVOID
GENERALIZING SYNERGY
ACROSS ALL DOMAINS



RECOGNIZE
HETEROGENEITY ACROSS
EXPERIMENTS



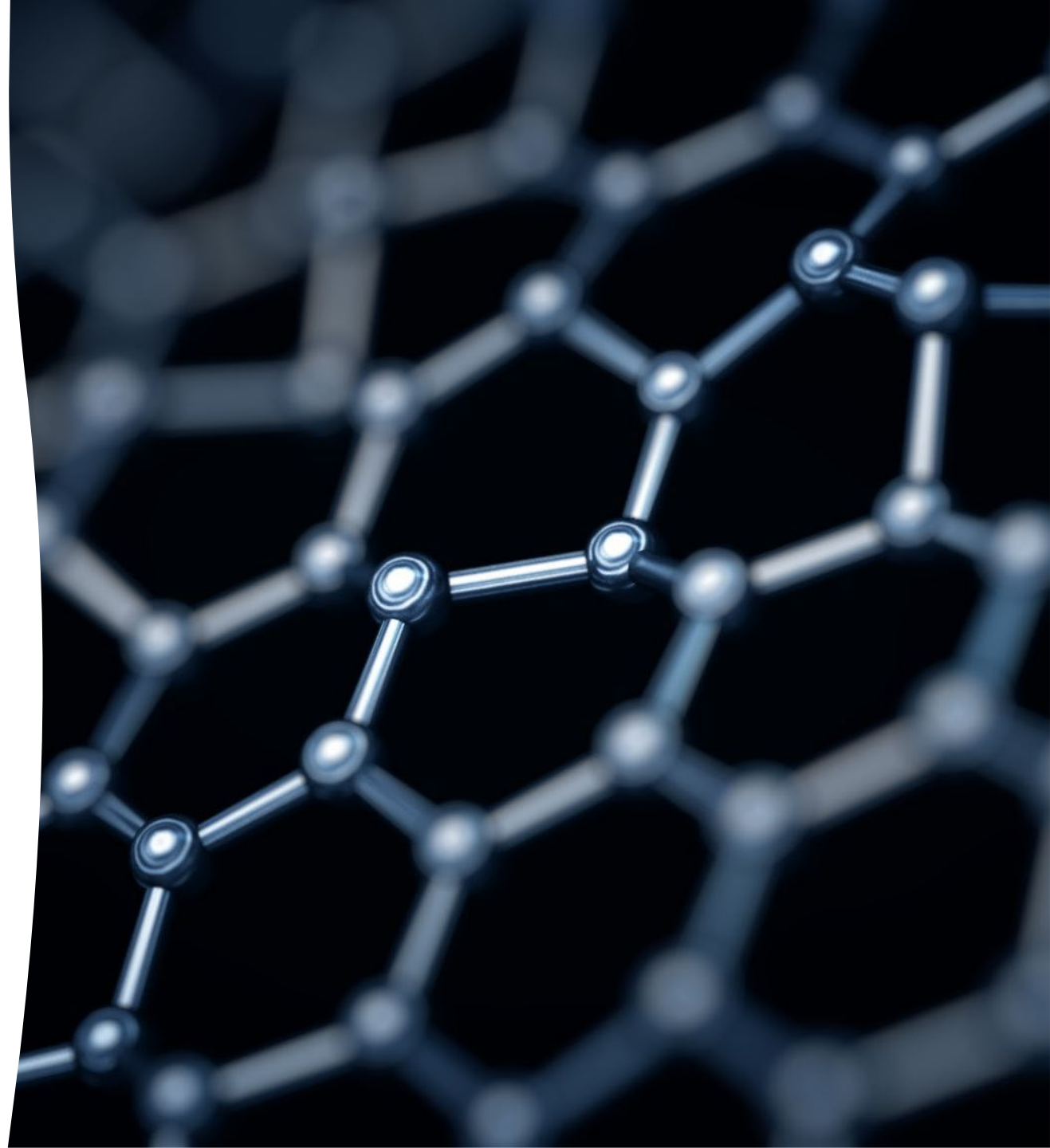
**WHAT NEEDS TO BE ACCOUNTED FOR
TO CREATE HAI COLLABORATIONS**

Behavioral & Physiological Effects of AI Teammates

Qin et al. (2025): VR sensorimotor task — human+AI underperformed human-only at high difficulty

Findings: elevated arousal, lower engagement, reduced human-to-human communication

Implication: AI insertion can disrupt human dynamics



Trust, Monitoring, & Gaze Signals



Ries et al. (2024): Overcooked AI experiments — eye gaze, trust, coordination metrics



Agents adapting to human gaze achieved better trust and coordination



Gaze patterns correlated with trust and attention distribution

Complementarity & Role Allocation

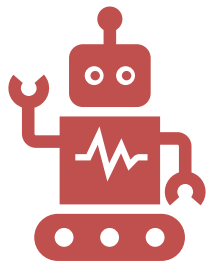
Hemmer (2025):
maximize
complementary
strengths

Humans: ethics,
context

AI: scale,
consistency

Avoid redundancy
or overlap to
reduce conflicts

Mental Modeling & Risks



Chakraborti & Kambhampati (2018):
Should AI model human
beliefs and preferences?



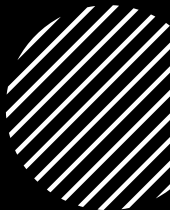
Risk: misaligned models can lead to
manipulation or deception



Ethical dimension: collaboration
embeds value judgments



Design Challenges & Open Problems



Trust calibration: avoid over/under trust



Transparency/explanation: how much, when, to whom



Shared mental models & alignment



Dynamic role adaptation, conflict resolution



Workflow integration & adoption barriers

Training & Onboarding in HATs



National Academies (2022): new training models needed



Approaches: cross-training, perturbation training, adaptive practice



Challenges: psychological differences, interface literacy

Risks & Pitfalls Recap



Over/under trust in AI



Misaligned objectives or communication



Interface overload, cognitive burden



Ethical risks: bias, misuse, autonomy issues



Organizational adoption resistance

Assignments Reminder & Weekly Reading



Submit Research Journal #1
(Vaccaro et al. meta-analysis)



Submit AI in the News #1
(example of collaboration)



Prepare for Week 2:
Understanding Different Human-
AI Patterns